Secular Changes, Polarity Epochs and Tectonic Movements as Indicated by Palaeomagnetic Studies of Hungarian Rock Samples[1])

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Summary – The paper contains an outline of the palaeomagnetic investigations in Hungary including some of their geophysical and geological applications. More than 500 samples of volcanic rocks were studied by usual palaeomagnetic techniques. The rocks investigated belong to three different geological periods – the Quaternary – upper Tertiary, the middle Miocene and the lower Cretaceous. The palaeomagnetic data obtained for the individual localities are summarized in the Appendix.

Introduction

The aim of the paper is to give an outline of palaeomagnetic investigations in Hungary.

The results of the palaeomagnetic studies of the Plio-Pleistocene basalts, middle Miocene andesites, and lower Cretaceous volcanic rocks, collected from the sites drawn in Fig. 1, are presented. Palaeomagnetic investigations of volcanic rocks of similar age in the countries surrounding Hungary were carried on by the following authors: BUCHA and others, NAIRN, in Slovakia, (BUCHA and others [1][4]), NAIRN [13, 14]), MIKHAYLOVA and GLEVASSKAYA, in Soviet-Transcarpathians (MIKHAYLOVA and GLEVASSKAYA [12]), COSTA-FORU (with co-workers), in Roumania (COSTA-FORU and others [2]).

Methods

The method of sampling and measuring adopted was similar to the one used commonly. From a single outcrop if the conditions allowed at least six samples were collected. Out of each sample two cubes with an edge of 2.5 cms were cut. Beyond the natural remanent magnetization, the induced magnetization in the earth's magnetic field was also measured. Throughout a sample group the Koenigsberger ratio Q_n was in general of the same value in order of magnitude and an anomalously small Q_n-value indicated unstable magnetization.

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[3]) Numbers in brackets refer to References, page 162.

At the beginning of our palaeomagnetic studies the direction and intensity of magnetization was measured using a rock generator of low sensitivity. (MÁRTON and others [7]). Currently, several types of astatic magnetometers are used with sensitivities up to 0.06 γ/scale division.

As a routine technique each sample is subjected to progressive demagnetization in alternating fields of up to several hundreds of oersteds in steps of 100 oersteds, the magnetization being determined after each step. Site mean direction of magnetization is computed for the demagnetization stage where scatter estimated visually on the stereographic projection is a minimum is adopted as the best representation of the original magnetization. The mean direction of magnetization and its statistical parameters are computed following the method of FISHER (FISHER [4]).

Besides the measurements of magnetization thermomagnetic curves are obtained for estimating the Curie temperatures. Polished sections are also examined using an ore microscope. At least one sample per locality is prepared for these examinations. Results of these, however, will not be included here.

Plio-Pleistocene basalts

In Hungary, remains of the postorogen basaltic volcanism of Plio-Pleistocene age can be found in Transdanubia near Lake Balaton and at the northern foreground of the Mátra Mountains, North-Nógrád district, respectively (Fig. 1), having a con-

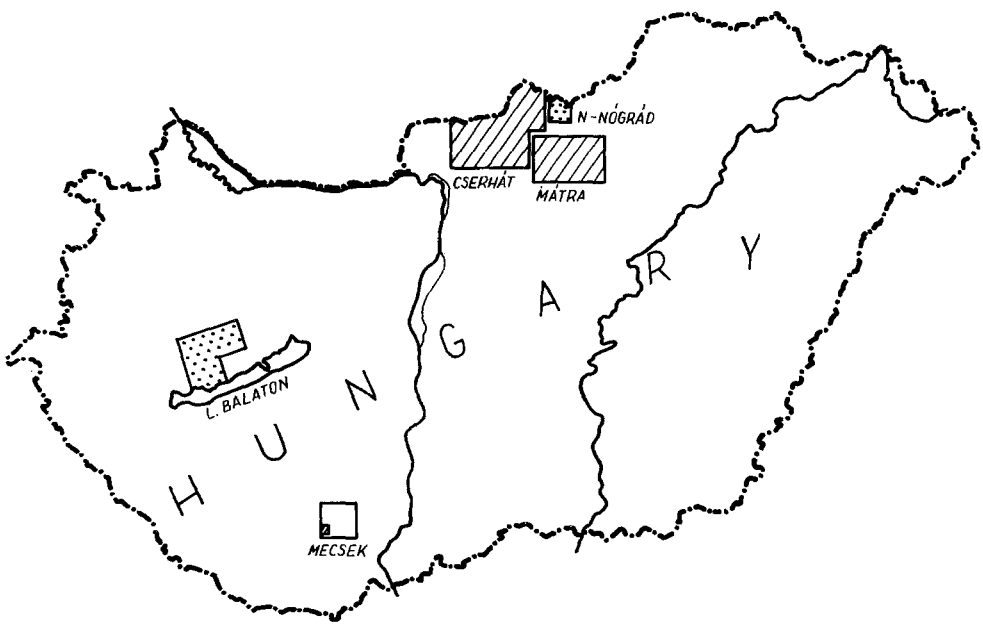


Figure 1
Map showing distribution of sampling districts. Legend: postorogenic basalts; middle Miocene andesites; lower Cretaceous volcanic rocks

tinuation also in Slovakia. In both areas the late volcanoes occur in form of volcanic cones, dykes, necks and lava plateaus. Formations underlying the basalts in the various localities are sedimentary rocks of Triassic, Oligocene, and upper Pliocene age. In Slovakia, the older basaltic lava overlies tuffites with Montpelier fauna an age indicating deposit. Either in Transdanubia or in the North-Nógrád district, no chronological order in the individual eruptions can be geologically established on account of, partly the wide variety of the underlying rocks, partly, the spatial separation of the volcanic products.

Of 16 localities in Transdanubia and of 5 localities in the North-Nógrád district. 12 resp. 3 ones have stable thermoremanent magnetization. Of the Transdanubian localities 6 exhibit normal magnetization. The remainders and all the North-Nógrád

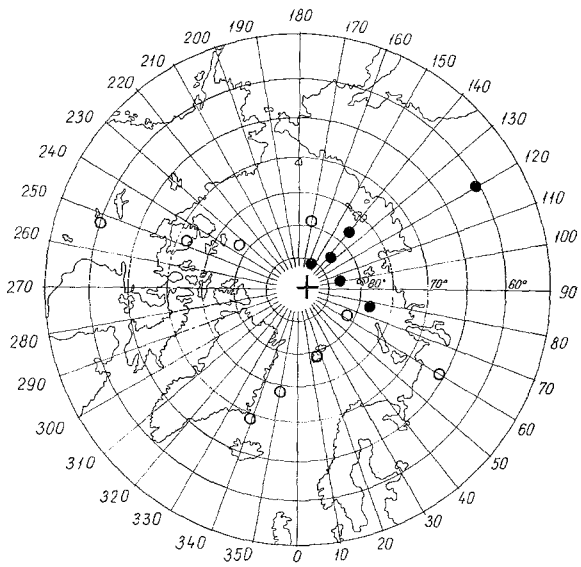


Figure 2
Positions of virtual geomagnetic poles calculated from site mean vectors of magnetization for Plio-Pleistocene basalts. Position of mean VGP is also included. Legend: ● full circle: geomagnetic north pole; ○ open circle: geomagnetic south pole; +: mean VGP

localities are reversely magnetized. Relevant palaeomagnetic data are summarized in the Appendix. As the dislocations affected the basalts, have been at most confined to slight vertical displacements the scatter of VGPs (Fig. 2) calculated from the site mean directions can be interpreted as a result of the secular variation of the earth's magnetic field.

In Slovakia, in the immediate vicinity of North-Nógrád district, the oldest member of the post-orogenic basalts is reversely magnetized (NAIRN [13]), whilst the younger exhibits normal magnetization. As regards to this result and to the shortness of this volcanic period (an observational result which takes into account the quantity of

volcanic material and the violence of eruptions), the reversely magnetized basalts must be older in Hungary, too. More precisely, the lavas of basalt under discussion flew out in the period immediately preceding and succeeding the last field reversal 0.7 my. ago (COX and DALRYMPLE [3]).

Middle Miocene andesites

The products of the orogenic phase in the inner Carpathian volcanism can be traced from the lower Helvetian to the lower Sarmatian in the volcanic sequence of the Mátra Mountains, North Hungary (Fig. 1). The lava flows are predominantly represented by the andesite suite which can be divided into three parts: lower, middle and upper andesite. The age of the lower andesite is upper Helvetian and the middle and upper andesites belong to the Tortonian. The middle andesite horizon can still more precisely be divided into six subhorizons (KUBOVICS [6]), within which the chronological order is known. The andesites of the Cserhát Mountains (bordering on the Mátra Mountains from East) are placed in the Tortonian. They can petrologically be identified with the middle andesite horizon in the Mátra Mountains.

For palaeomagnetic studies samples were collected from 27 localities in the Mátra Mountains and from 5 localities in the Cserhát Mountains. The sample groups collected, apart from one rhyolite group, consisted of various kinds of andesites. After magnetic cleaning the directions of remanent magnetization of 23 localities of the 27 in the Mátra Mountains, and those of all 5 localities in the Cserhát Mountains can be taken as the directions of original TRM. Virtual geomagnetic pole positions are shown in Fig. 3. Relevant palaeomagnetic data are summarized in the Appendix.

The table of the succession of zones of magnetization (Table 1) was compiled from palaeomagnetic data so far obtained in the Mátra and Cserhát Mountains.

A portion of the NAIRN's table of polarity zones found by him from the palaeomagnetism of Slovakian volcanics is also included. According to the table the Tortonian, almost as a whole, is characterized by a negative palaeomagnetic zone. A positive zone in the lower Tortonian may probably be extended to the upper Helvetian. The positive polarity in the lower Tortonian is a new event as compared to the Nairn's scheme. To the lower Sarmatian a positive zone belongs. The only point which can be subjected to discussion is the correct datum of the uppermost polarity change, that is, it happened still in the upper Tortonian or on the Tortonian-Sarmatian boundary. The upper andesite the youngest member of the andesite volcanism in the Mátra Mountains, is believed as upper Tortonian by most of geologists. If this opinion were accepted, the polarity change would fall still into the Tortonian. On the other hand, Nairn proposed a correlation with New-Zealand and Japan as a working hypothesis (NAIRN [14]). Then, the polarity change in question would be placed on the Tortonian-Sarmatian boundary. By this means the upper andesite in the Mátra Mountains ought to be placed in the lower Sarmatian and this would not be inconsistent with the Mátra Mountains geology.

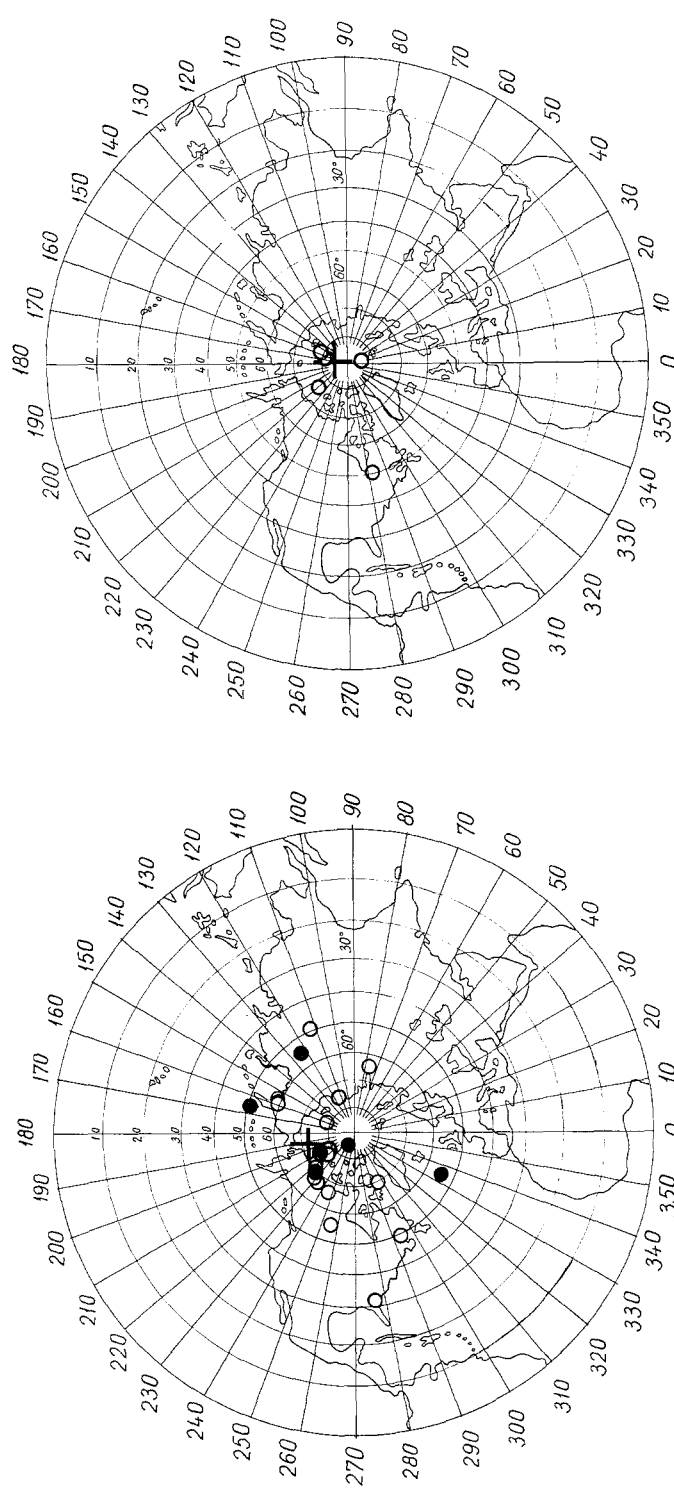


Figure 3

Positions of virtual geomagnetic poles calculated from site mean vectors of magnetization for Mátra Mountains andesites (except one rhyolite) (left) and for Cserhát Mountains andesites (right). Positions of mean VGPs, calculated excluding localities of which tectonic position have not been precisely determined, are also included. Legend as in Figure 2.

Table 1

Age and possible correlation of the Hungarian and Central Slovakian volcanic rocks

Mátra-Cserhát Mountains					Central Slovakia (Nairn, [13, 14])			
		Polarity	Palaeomagnetic zone	Polarity				
1. Sarmatian	rhyolite upper andesite	+ +	N	+ + +	3nd phase andesites intrusives 2nd phase andesites	Lower	Sarmatian	
Tortonian	middle andesite dykes VI	−	R	−	2nd phase rhyolites	Upper Middle	Tortonian	
	V	−						
	IV	−						
	III							
	II	−						
	I	+	N		1st phase andesites	Lower		
u. Helvetian	lower andesite	+?						

Lower Cretaceous volcanic rocks

In this part of the paper the palaeomagnetism of 12 lower Cretaceous and one Tertiary volcanic rock outcrops in the Mecsek Mountains, Southern Hungary, is concerned (Fig. 1).

The sampling area (Eastern Mecsek, Zengő range) is a perisyncline with a sedimentary sequence, penetrated by phonolite and trachidolerite lavas, ranging from the Liassic to the middle Cretaceous. Both phonolite and trachidolerite eruptions took place in connexion with the late part of the Neo-Cimmerian phase of orogeny and they are placed in the Valanginian and Hauterivian stages of the lower Cretaceous. The age of the amphibole andesite (No. 67 sampling site, see Appendix) is likely, Helvetian.

Of the cycles of dislocations that affected the deposits of the Zengő range the Austrian phase developed the principal elements (foldings) and the Rhodanian phase brought about imbrications, overthrusts and normal faults.

Of the 13 sampling sites 7 lower Cretaceous outcrops and the Helvetian one exhibit stable magnetization. After the corrections for tilt had been applicated 6 Valanginian localities showed reverse magnetization, whereas a lower Cretaceous trachidolerite dyke and the Helvetian andesite were normally magnetized. Relevant palaeomagnetic data are summarized in the Appendix. It is noticeable that all Valanginian localities are reversely magnetized that is the polarity of the earth's magnetic field was opposite to that of the present field during the main phase of the lower Cretaceous volcanism.

Irrespective of polarity the site mean directions of magnetization cluster about two-points: 1; $D_1 = 349.3°$, $I_1 = 51.2°$ and 2; $D_2 = 83.1°$, $I_2 = 62.0°$, respectively (Fig. 4). The deviation in D is significant. The values of D_1, I_1, however, agree well with those of $D_E = 6.9°$ $I_E = 54.9°$ the site mean for the Mecsek Mountains obtained from the mean Eurasian Cretaceous VGP (MÁRTON and SZALAY [10]). The deviation of the site mean D_2, I_2 from D_1, I_1 (or what amounts to the same from the site mean D_E, I_E calculated using the mean Eurasian Cretaceous VGP) suggests a clockwise rotation by about 90°, about a vertical axis of the localities having the mean direction D_2, I_2.

This can be readily correlated with the involved changes of strike in the Mecsek Mountains geology. The rotations about a vertical axis revealed by the palaeomagnetic data have the same general tendency as the deviations in strike from the folding axis of the anticline containing the outcrops of Group 1. These deviations in strike can be read off the geological map at the sampling localities of Group 2. The rotations determined by palaeomagnetic means interpret at the same time the deviations in strike as a result of the rotation components of the overthrusts still preceded by the intrusion of the Helvetian andesite.

Conversely, the deviations in strike measured at the localities of Group 2 give a further possibility of correcting the palaeomagnetic site mean vectors of these localities. Rotation corrections were established for the individual sites of Group 2 from

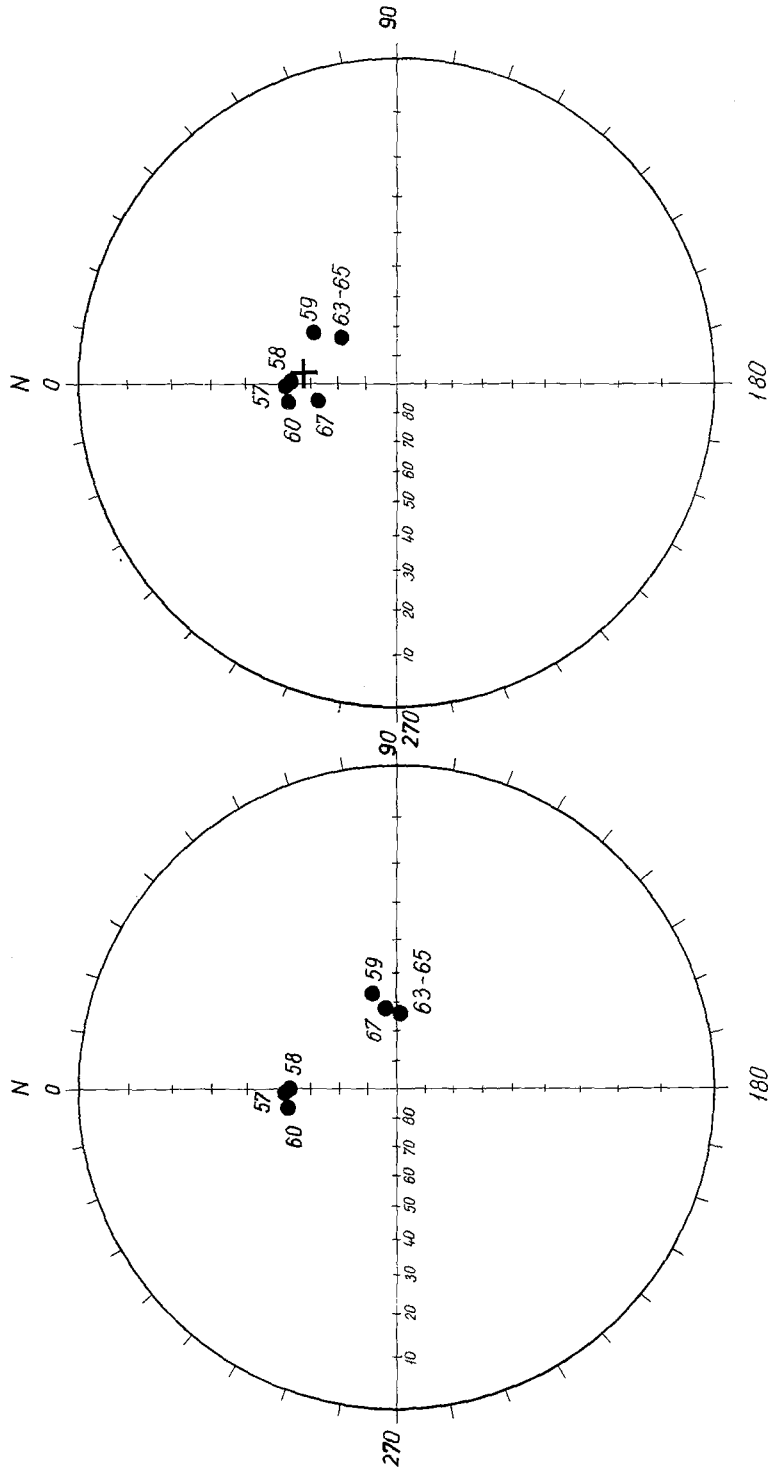


Figure 4
Palaeomagnetic site mean directions, converted to normal magnetization for localities of Mecsek Mountains. Left: site means after tilt correction. Right: site means after rotation correction for localities of 59, 63–65, 67. Mean direction of site mean vectors of magnetization are also shown. (Locality 67 excluded because of its tertiary age)

the geological map and $\phi = 81.6°$; $\Lambda = 160.3°$ with statistical parameters $\alpha = 20.6°$ and $\kappa = 15$ was obtained for the position of the mean lower Cretaceous VGP.

Appendix. Palaeomagnetic data

Plio-Pleistocene basalts

Localities	N/N_0	φ^0	λ^0	$\bar{D}^0$	$\bar{I}^0$	κ	α^0	ϕ^0	Λ^0		
Transdanubia											
1. Uzsa, Szebike	7/8	46.9	17.3	14.1	56.1	17	15.1	77	140		
2. Sümeg	5/5	46.9	17.3	3.8	61.8	100	7.6	86	158		
3. Tátika	5/6	46.9	17.3	17.6	68.6	100	7.6	78	76		
4. Zalaszántó	6/6	46.9	17.3	8.5	61.8	125	6.1	83	137		
5. Vindornyaszöllös	7/7	46.9	17.3	35.6	53.3	9	21.6	60.5	120		
6. Hegyestü	5/5	46.9	17.7	9.8	65.2	36	12.9	83	100		
7. Gulács	9/10	46.8	17.5	164.6	−74.4	67	6.5	−74	170		
8. Badacsony	7/9	46.8	17.5	153.1	−74.5	54	8.2	−69	159		
9. Sághegy	4/4	47.2	17.1	179.0	−72.9	75	10.5	−79	196		
10. Haláp	10/13	46.9	17.5	187.0	−56.3	50	6.9	−79	350		
11. Kabhegy	12/14	47.0	17.6	170.7	−56.1	69	5.2	−78	54		
12. Szentgyörgyhegy	5/5	46.1	17.7	153.5	−53.4	80	8.6	−72	72		
13. Bazsi	0/6				excluded						
14. Diszel	0/4				unstable						
15. Tótihegy	0/9				unstable						
16. Sümegprága	0/6				unstable						
North-Nógrád											
17. Medves	55/62	48.2	19.9	154.1	−44.9	34	3.5	−60	72		
18. Pécskő	12/12	48.1	19.8	218.1	−70.1	55	5.9	−66	240		
19. Szilváskő	7/7	48.1	19.9	191.6	−62.3	35	10.2	−82	244		
20. Nagykő	0/2				excluded						
21. Nagy Salgó	0/7				excluded						
22. Kis Salgó	0/5				excluded					κ	α^0
Mean VGP calculated from VGP of 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 17, 18, 19								88.2	96.8	23	8.4

Mátra Mountains

Localities	N/N_0	φ^0	λ^0	$\bar{D}^0$	$\bar{I}^0$	κ	α^0	ϕ^0	Λ^0
23. Ágasvár, lower andesite	0/6	stable, self reversed, excluded							
24. Csörgőpatak, middle andesite, horizon I.	6/6	47.9	19.8	299.3	+74.9	70	7.7	+54.5	335.0
25. Óvár, middle andesite, horizon I.	6/6	47.9	19.8	356.9	+52.1	83	7.4	74.6	209.7
26. Báránykő, middle andesite, horizon I.	3/7	47.9	19.8	349.8	+48.3	200	8.7	69.8	226.5
27. Tippanos, middle andesite, horizon II.	6/8	47.9	19.8	203.0	−40.4	15	17.8	−59.2	335.2

Localities	N/N_0	φ^0	λ^0	$\bar{D}^0$	$\bar{I}^0$	κ	α^0	ϕ^0	Λ^0
28. Hasznosi Várhegy, middle andesite, horizon II.	3/6	47.9	19.8	175.3	−55.8	200	9.8	−77.9	38.2
29. Mátrakeresztes, middle andesite, horizon V.	6/6	47.9	19.8	168.1	−46.9	83	7.4	−68.2	49.1
30. Szurdokpüspöki I, middle andesite, horizon VI.	6/8	47.8	19.7	150.3	−63.8	33	11.8	−69.6	114.6
31. Mátrakeresztes, middle andesite, horizon VI.	7/7	47.9	19.8	196.4	−37.0	67	7.5	−59.8	348.1
32. Nagylapát, middle andesite, horizon VI.	6/6	47.8	20.0	165.6	−48.5	100	6.8	−68.6	56.4
33. Mogyorósorom, middle andesite, horizon VI.	6/6	47.9	20.0	189.5	−58.2	56	9.0	−78.7	339.1
34. Tar, middle andesite	3/3	47.9	19.8	120.3	−28.4	29	23.5	−31.5	98.0
35. Szurdokpüspöki II, middle andesite, horizon VI.	0/32				unstable				
36. Gyöngyöspata, middle andesite	6/6	47.8	19.8	355.2	+69.0	250	4.2	84.4	348.3
37. Nagybátony, dyke I, middle andesite	5/5	48.0	19.8	172.4	−48.3	44	11.6	−70.4	39.9
38. Nagybátony, dyke II, middle andesite	3/3	48.0	19.8	178.2	−48.3	40	19.7	−71.2	24.7
39. Tar, dyke, middle andesite	3/4	47.9	19.8	230.4	−50.5	15	32.5	−48.9	294.5
40. Csákánykő, dyke, middle andesite	6/6	47.9	20.1	176.9	−56.8	83	7.4	−79.3	33.5
41. Fényespusztai Várhegy, dyke, middle andesite	4/7	47.9	20.0	128.3	−53.4	60	12.0	−49.4	111.5
42. Máté keresztje, dyke middle andesite	6/6	47.9	20.1	154.1	−48.4	120	5.5	−62.8	76.5
43. Rudolftanya, upper andesite	5/5	47.9	20.0	19.2	+63.3	133	6.6	76.4	116.9
44. Nyesettvár, upper andesite	0/12				unstable				
45. Kékes, upper andesite	0/12				unstable				
46. Nagyátalkő, upper andesite	6/6	47.9	19.9	39.9	+50.0	100	6.7	55.3	124.5
47. Sombokor I, middle Miocene andesite	6/6	47.9	20.0	220.5	−73.6	42	10.5	−64.0	252.5
48. Sombokor II, middle Miocene andesite	3/6	47.9	20.0	160.0	−48.7	50	17.6	−66.2	67.4
49. Gyöngyössolymosi Kishegy, lower Sarmatian rhyolite	6/6	47.8	19.9	21.8	+24.4	100	6.7	50.6	165.1
									κ α^0
Mean VGP calculated from VGP of 25, 26, 27, 28, 29, 31, 32, 33, 37, 38, 40, 42, 43, 46, 48, 49								73.1	194.2 23.4 7.8°

Cserhát Mountains

Localities	N/N_0	φ^0	λ^0	$\bar{D}^0$	$\bar{I}^0$	κ	α^0	ϕ^0	Λ^0		
50. Berceli hegy, middle andesite, horizon III.	8/8	47.9	19.4	174	−54	58	7.2	−76	40		
51. Szanda hegy, middle miocene, horizon III.	8/8	47.9	19.4	190	−58	50	7.9	−79	336		
52. Kisgéc, middle andesite	7/11	48.1	19.6	177	−70	43	9.2	−84	184		
53. Zsuny, middle andesite	5/5	48.0	19.6	187	−60	100	7.7	−80	345		
54. Baráthegy, middle andesite	8/10	47.9	19.6	132	−48	39	9.0	−50	102	κ	α^0
Mean VGP calculated from VGP of 50, 51, 52, 53								83.8	183.5	60	12

Mecsek Mountains
φ=46.1°, λ=18.4°

Localities	N/N_0	$\bar{D}^0$	$\bar{I}^0$	κ	α^0	ϕ^0	Λ^0		
55. Zengővárkony, Valanginian trachidolerite, pillow lava	0/5				unstable				
56. Márévári völgy, Valanginian trachidolerite	0/9				unstable				
57. Márévári völgy, lower Cretaceous trachidolerite dyke	4/5	350.4	+51.5	27	18.0	75.5	225.0		
58. Márévári völgy, Valanginian trachidolerite	4/5	227.7	−60.1	60	12.0	77.5	196.0		
59. Hosszuhetény, lower Cretaceous trachidolerite dyka	5/6	210.5	−56.0	36	12.8	66.5	112.5		
60. Jánosi Nagyvölgy, Valanginian trachidolerite	6/6	205.4	−73.3	50	9.6	69.0	254.0		
61. Jánosipuszta, Hauterivian trachidolerite	0/2				unstable				
62. Márévári völgy, Hauterivian trachidolerite	0/7				unstable				
63. Mázai völgyfő, lower Cretaceous phonolite	3/3								
64. Mázai völgyfő, lower Cretaceous teshenite	2/2	219.0	−64.5	17	13.9	63.5	92.0		
65. Mázai völgyfő, lower Cretaceous phonolite dyke	3/6								
66. Kövestető, lower Cretaceous phonolite	0/9				excluded				
67. Komló, Helvetian andesite	10/10	347.4	+61.8	112	4.2	79.0	258.0	κ	α^0
Mean VGP calculated from VGP of 57, 58. 59, 60, 63, 64, 65						81.6	160.3	15	20.6

Legend: N/N_0; samples used/collected
φ^0, λ^0; coordinates of site
$\bar{D}^0$; site mean declination } of magnetization
$\bar{I}^0$; site mean inclination }
κ; precision parameter
α^0; circle of confidence (95%)
ϕ^0; latitude } of VGP
Λ^0; longitude }

Note: Localities having tectonic position not exactly determined are neglected in calculation of mean VGP.

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